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GTXplorer: A portal to navigate and visualize the evolutionary information encoded in fold A glycosyltransferases

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Abstract

Glycosyltransferases (GTs) play a central role in sustaining all forms of life through the biosynthesis of complex carbohydrates. Despite significant strides made in recent years to establish computational resources, databases and tools to understand the nature and role of carbohydrates and related glycoenzymes, a data analytics framework that connects the sequence-structure–function relationships to the evolution of GTs is currently lacking. This hinders the characterization of understudied GTs and the synthetic design of GTs for medical and biotechnology applications. Here, we present GTXplorer as an integrated platform that presents evolutionary information of GTs adopting a GT-A fold in an intuitive format enabling in silico investigation through comparative sequence analysis to derive informed hypotheses about their function. The tree view mode provides an overview of the evolutionary relationships of GT-A families and allows users to select phylogenetically relevant families for comparisons. The selected families can then be compared in the alignment view at the residue level using annotated weblogo stacks of the GT-A core specific to the selected clade, family, or subfamily. All data are easily accessible and can be downloaded for further analysis. GTXplorer can be accessed at https://vulcan.cs.uga.edu/gtxplorer/ or from GitHub at https://github.com/esbgkannan/GTxplorer to deploy locally. By packaging multiple data streams into an accessible, user-friendly format, GTXplorer presents the first evolutionary data analytics platform for comparative glycomics.

Key words: evolution, glycosyltransferases, phylogeny, visualization, web tool

Introduction

Glycosyltransferases (GTs) catalyze the biosynthesis of complex carbohydrates, one of the most abundant biopolymers present in any cell. Defects in GTs have been implicated in several diseases such as congenital disorders of glycosylation, neurodegeneration and cancer (Clarke et al. 2012; Kinoshita et al. 2014; Joshi et al. 2018; Gupta et al. 2020; Moll et al. 2020). In addition, there is a significant interest in the optimization, synthesis and engineering of GTs for applications in biofuels, food products, agricultural and livestock industries, and more recently, for the production of a new generation of highly efficient glycoconjugate vaccines (Himmel et al. 2007; McArthur and Chen 2016; Micoli et al. 2019). GTs are currently classified into 114 families by the CAZy database, most of which adopt one of three major folds: GT-A, B or C (Lombard et al. 2014). The diversity in their primary sequences with large insertion loops in the catalytic domain and the presence of very few conserved catalytic residues have presented a significant challenge in comparative analyses across this large family of enzymes. Recently, we relied on the structural

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conservation within the GT-A fold families to present a comprehensive evolutionary framework connecting the functional diversity across all GT-A fold enzymes (Taujale et al. 2020). This study has generated a wealth of curated information such as curated alignments of over half a million GT-A sequences, phylogenetic profiles and annotations for statistically conserved amino acid positions that can provide key insights into the relationships between sequence, structure, function and regulation of GT-A function through comparative sequence analyses and serve as a conceptual starting point for generating testable hypotheses. However, the general usability and interpretability of such complex data structures hinges on presenting these contextual information through a coherent user-friendly visualization scheme.

The past decade has seen much progress toward bioinformatics resources for glycoscience research. For example, GlyGen (York et al. 2020) provides an integrative portal to retrieve glycoconjugate-related data. Multiple glycoenzymes-related databases provide curated interlinked information about their primary sequences (Lombard et al. 2014; Egorova et al. 2021; Yamada et al. 2020). Resources focusing on the structural aspects of glycoscience, specifically the conversion of carbohydrate sequences to 3D models (Kirschner et al. 2008; Engelsen et al. 2014), cataloging the 3D structural information of carbohydrates and related enzymes (Pérez et al. 2015a) and visualization of carbohydrate structures (Sarkar and Pérez 2012:3, Pérez et al. 2015b) have already proven to be immensely valuable in driving glycoscience research. However, in order to make sense of the layered information encoded in the expansive range of different glycans, we first need to understand the evolution of biosynthetic machinery that led to this diversification. Yet, among these budding compendiums of resources, most focus on the glycans and not on their biosynthetic machinery, while others lack a data analytics framework for mining the relationships connecting sequence, structure, function and evolution.

Here, we present GTXplorer, an interactive tool for the visualization of GT-A evolutionary information which provides an integrated platform for comparative analysis of more than half a million sequences that adopt a GT-A type fold and are spread across 9 clades, 53 families and 99 subfamilies. By presenting complex data sources linking the evolutionary relationships between these diverse families of enzymes in an intuitive browser, GTXplorer enables researchers to extract and use the evolutionary data encoded in thousands of GT-A sequences from diverse organisms to interpret and predict the functional impact of natural variants in primary sequences. GTXplorer offers modular access to perform family, subfamily or amino acid residue level comparisons through an interactive phylogenetic tree navigator and an annotated contrast alignment viewer. Built using the React JS and JavaScript libraries, users can easily navigate, view and download relevant information for generating publication-quality images and supporting other in-house analyses. GTXplorer connects data from disparate sources and formats into an interpretable form that is readily accessible as a web portal or a locally built application for performing comparative evolutionary analyses of GT-A fold enzymes and generating novel testable hypotheses driving experimental research in glycoscience.

Results

The GTXplorer portal

We present GTXplorer as a user-friendly interactive portal to navigate large volumes of data related to the GT-A fold families and subfamilies. It includes a diverse portfolio of information related to the classification and distribution of GT-A sequences, along with a repository of sequence, alignment and phylogenetic data (Figure 1).

Data is organized into three layers of GT-A classification (clades, families and subfamilies) as outlined in (Taujale et al. 2020). The topmost level is the nine major GT-A clades that include families, some of which are further subdivided into subfamilies based on conserved pattern positions. Two modes of navigation options are provided: 1) A tree view mode which provides an interactive phylogenetic tree that users can click on to access detailed information about specific GT-A clades and families and 2) An alignment mode where users can access a hierarchical layout of GT-A clades, families and subfamilies that can be selected to generate comparative alignments using weblogos (Crooks 2004). Both views offer insights into different aspects of GT evolution. The phylogenetic tree depicts the evolutionary relationships across GT-A families. Clicking on the clades or families in the tree view brings up a card that includes a short description of the known roles of sequences within that clade/family along with additional information about the mechanism, domain organization, sub-classification and quantified presence across major taxonomic groups.

On the other hand, the alignment view offers a way to compare large sequence alignments across clades/families/subfamilies to identify conserved positions at different hierarchical levels. The alignment is annotated with conserved structural features, motifs and other regions of functional importance. Any sequence from the hierarchical level can be used to number the aligned position, thus allowing easy mapping of these features to any sequence of interest. Also, we annotate the conserved pattern positions identified in our previous Bayesian pattern-based analysis for all the subfamilies, which provide informed targets for mutational analyses in functional studies (Taujale et al. 2020) (gray bars in the bottom panel of Figure 1). Users can use the search bar to enter a clade/family/subfamily name or the UniProt ID to only show the families that are of interest.

These intuitive view modes with multiple customization options can also help generate publication quality figures. The datasets are generated from two large sequence repositories (NCBInr (Pruitt et al. 2007) and UniProt proteomes (UniProt Consortium 2019)), and download links are conveniently placed throughout the navigation pages to download any specific piece of information desired from any of the two repositories. Users can download raw full-length sequences in fasta format for any particular clade/family/subfamily. In addition, users can also download fasta files with an alignment of the GT-A domains or tabular formatted files with additional information.

Example case studies with GTXplorer to infer catalytic mechanisms and donor specificity

We illustrate the use of GTXplorer in hypothesis generation using two case studies outlining the observations made in (Taujale et al. 2020). For the first case, we observe the natural variation in the catalytic base position (xED-Asp) in the context of multiple evolutionary origins of retaining and inverting mechanisms (Figure 2A). GT-A families with an inverting mechanism use a catalytic base to deprotonate the acceptor nucleophile, facilitating the initiation of a nucleophilic attack with direct displacement of the phosphate leaving group (Lairson et al. 2008; Gloster 2014). In contrast, for the retaining GT-As, an alternative SNi mechanism has been proposed where the phosphate leaving group acts as the catalytic base instead (Moremen and Haltiwanger 2019). This is in light of crystal structures with acceptor complexes that lack a suitably positioned conserved residue

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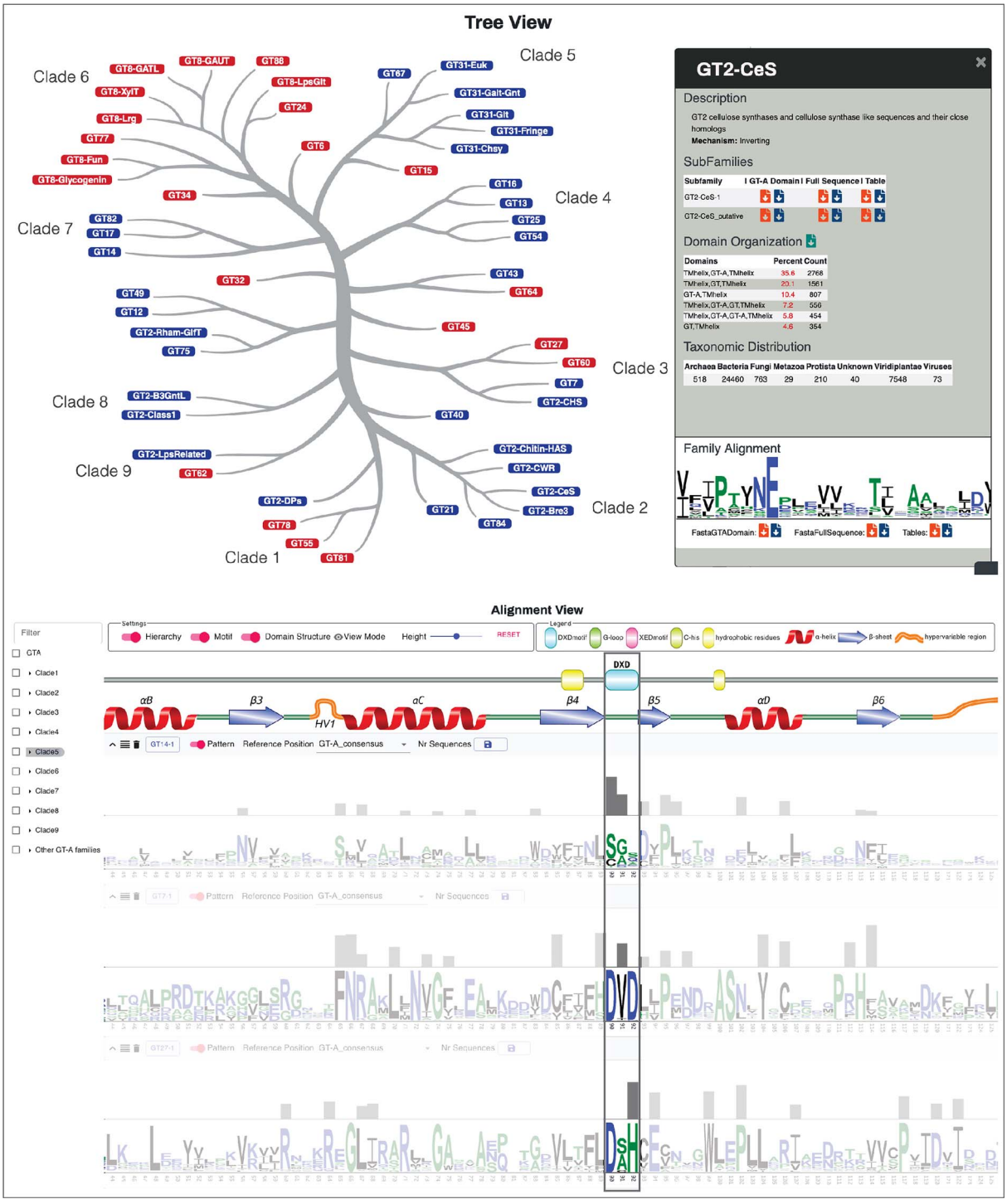


Fig. 1. The GTXplorer web interface. Top: The tree view panel shows the interactive tree on the left. Upon clicking a node (in this case, family GT2-CeS), a card appears on the right with related information about that node. Bottom: A snapshot of the alignment view highlighting the variation in the DXD motif between the metal independent GT14-1 subfamily and the metal-dependent GT7-1 and GT27-1 subfamilies.

that could act as a catalytic base supported by mutational studies. Thus, the two mechanisms exert different evolutionary pressures on the catalytic base position, which would have to be conserved in the inverting enzymes since it is essential for catalytic activity. This can be visualized using GTXplorer where a user could: Step 1: Find and select families from inverting or retaining clades (for example, inverting families GT2-CWR, GT21 and GT16 from clades 1 and 6 and retaining families GT8-GAUT and GT88 from clade 9) in the phylogenetic tree and the alignment viewer, Step 2: Turn on the radio button for motif annotation and Step 3: Scroll to find the xED motif position in the alignment viewer. In the alignment, the conservation of the xED-Asp is clearly highlighted for the selected

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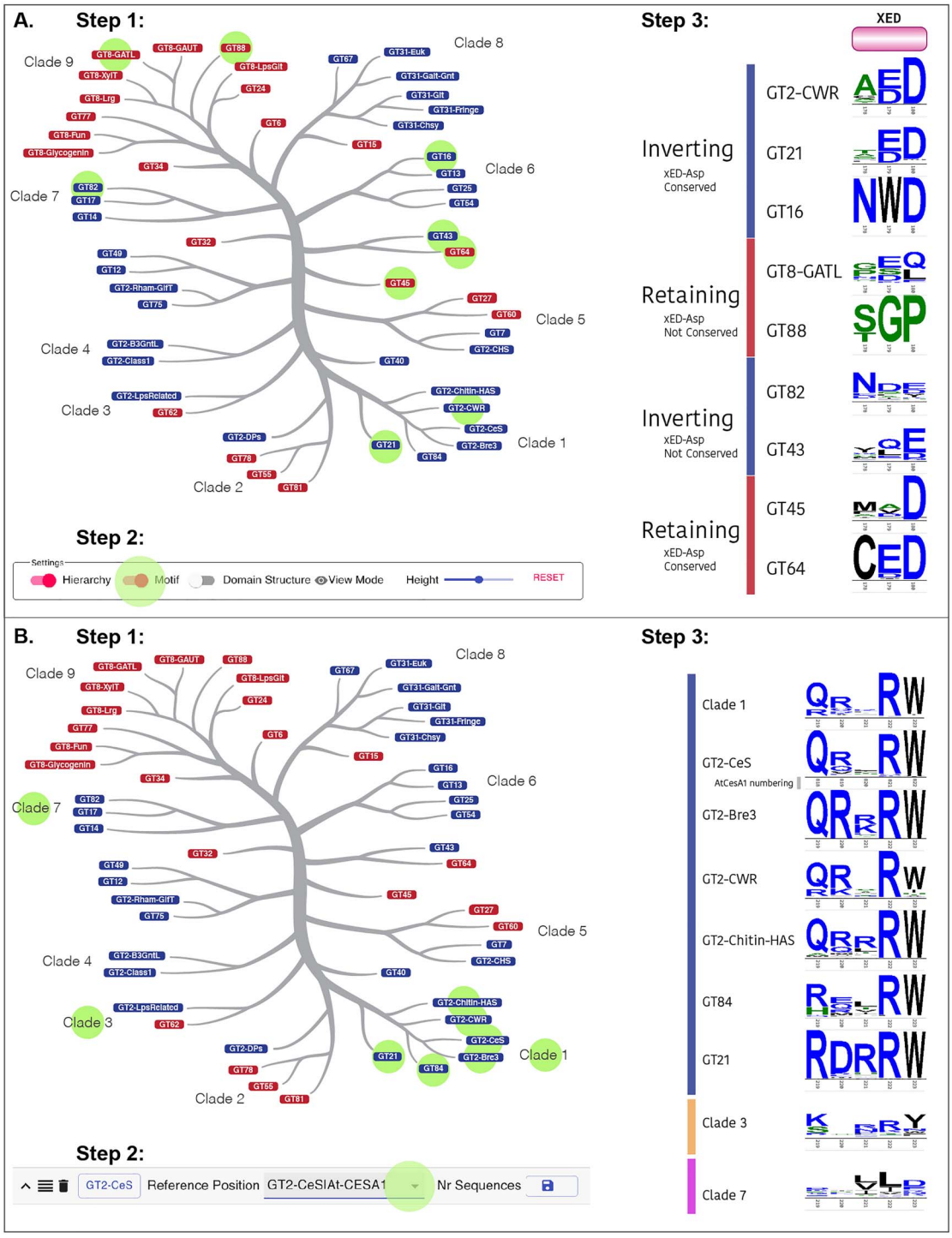


Fig. 2. Example cases for the use of GTXplorer. (A) Steps to visualize changes in the catalytic base position (xED-Asp) based on the mechanism of different families. (B) Steps to identify Clade 1 specific [QR]XXRW motif involved in donor and acceptor binding.

inverting families. However, in our previous study, we observed that there were a number of families that did not adhere to this rule where retaining families conserved the xED-Asp residue and inverting families like GT2-DPs, GT43, GT14 and others found alternative modes of inversion relieving the constraints to conserve the xED-Asp (Taujale et al. 2020). These exceptional families generally occurred in clades where inverting and retaining families were grouped together or closer to one another. This observation can be readily accessed in GTXplorer by simply selecting additional families from other clades (such as GT43, GT82, GT64 and GT45) and visualizing their alignment at the xED motif in the alignment viewer (Figure 2A). By making such comparative sequence analyses readily accessible, GTXplorer enables users to find trends and derive hypotheses based on the evolutionary conservation of specific residue positions.

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For the second case, we use GTXplorer to identify a clade-specific motif essential for substrate binding. Clade 1 groups processive GT-A families generally involved in the biosynthesis of large polysaccharides or glycosphingolipids. A [QR]XXRW motif has been identified in its members, where it has been shown to be important for binding both the donor and the acceptor (Morgan et al. 2013). We illustrated the extent of conservation of this motif within the Clade 1 members. GTXplorer allows users to conveniently locate such conserved motifs starting from their sequence of interest (Figure 2B). For example, a user can quickly start from a sequence of interest, like the *Arabidopsis thaliana* cellulose synthase 1 (AtCesA1), which is one of the subunits required for cellulose biosynthesis (Taylor et al. 2000). This sequence is part of the GT2-CeS family which can be identified from the table available for download through GTXplorer. By displaying the weblogo alignment for GT2-CeS by selecting that family in the phylogenetic tree and the alignment viewer, users can locate the conserved [QR]XXRW motif and even number the alignment positions using the AtCesA1 sequence as reference (Figure 2B Step 2). To make comparisons of this motif with other members of the clade, other families of Clade 1 can be selected, all of which show up in the alignment viewer and display the conserved motif. To contrast this with other clades, users could choose, for example, Clade 3 and 7 where these motif positions are not conserved at all, highlighting the conservation of [QR]XXRW motif only within clade 1 members. As we have demonstrated, GTXplorer can be a very useful tool in finding such evolutionarily constrained positions that point to conserved modes of function such as substrate binding.

Conclusion

The wide array of well-curated information and its accessibility provided by GTXplorer make it a user-friendly, interactive community resource that can strengthen computational and experimental research into the GT-A fold families, as highlighted by the specific examples provided. Coherent presentation of evolutionary information encoded in sequences allows experts and novices to easily visualize and perform comparative analyses of sequence conservation across evolutionary groups of GT-As. This analysis can then either be used to make interpretations of their results at hand or derive testable hypotheses to guide future experiments in glycoscience. As new GT-A fold families/sub-families are discovered, we plan to update them in GTXplorer and further refine the phylogenetic tree. We will also add structural information for cohesive access to phylogenetic, functional and structural data. We will also continuously improve and add new utilities to enhance interactive visualization and mining based on user feedback.

Methods

The interactive tree used in the tree view is generated using an annotated SVG file. Upon clicking on the tree nodes, users interact through a jQuery JavaScript code that pulls the relevant card along with the contained information.

The alignment view is built on a component-based framework using the React 17.0 library that provides flexibility, interoperability and reusability required by our data analytics platform to handle multiple data sources. GTXplorer is composed of three main components: 1) a hierarchical tree viewer for selecting clades, families and subfamilies, 2) the main comparative alignment viewer and 3) the settings section. The hierarchical tree is rendered by parsing a JSON file in the backend that includes associated information about the clades, families and subfamilies. The comparative alignment viewer displays summarized views of the alignment at the clade, family and sub-family levels rendered as weblogo images. These weblogos may be dynamically rearranged within the interface and provide options to download or display additional information. The settings section provides functions to customize the alignment view. This flexibility provides an implementation that can easily be adjusted using auxiliary files to accommodate new datasets or adapt the tool for visualizing an entirely different protein family.

The web version of GTXplorer is hosted on the Netlify cloud computing platform. A complete tutorial video for the use of the web tool can be accessed at https://vulcan.cs.uga.edu/gtxplorer/#Navigation. Users can also view, and download GTXplorer from GitHub and deploy a local version of the application using the provided instructions ensuring continued access to both the data and the interactive platform.

The GT-A phylogenetic tree, sequences, alignments and taxonomic information were obtained from (Taujale et al. 2020). The phylogenetic tree was redrawn in Adobe Illustrator for better esthetics. The alignment weblogos were generated using the Weblogo tool (version 3.7) (Crooks 2004). Domain bounds and domain organization information were collected by querying the GT sequences using Batch CD-Search (Marchler-Bauer et al. 2017). Full methods regarding generation of the sequence alignments and phylogenetic tree were detailed in a previous paper (Taujale et al. 2020). The sequences used to generate the weblogos for GTXplorer can be downloaded from https://doi.org/10.5061/dryad.v15dv41sh.

Availability

GTXplorer is freely available from GitHub at https://github.com/esbgkannan/GTxplorer, and the online web tool can be accessed at https://vulcan.cs.uga.edu/gtxplorer/.

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Conflict of interest statement

None declared

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